INCH - POUND

MIL-F-13198C (PA)
3 June 1975
SUPERSEDING
MIL-F-13198B (MU)
15 July 1971

MILITARY SPECIFICATION

FUZE, TIME, M65A1
METAL PARTS FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the fabrication of parts for one type of time fuze designated as Fuze, Time, M65A1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

DRAWINGS

PICATINNY ARSENAL

9207568 - Fuze, Time, M65A1

Beneficial comments, (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U. S. ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3. REQUIREMENTS

3.1 Material and Parts.-Material and parts shall be in accordance with the applicable drawings, standards and specifications.

3.2 Shear wire load.-The shear wire shall not shear when a load of less than 25 pounds is applied but must shear when a load of 35 pounds is applied.

3.3 First Article Inspection.-This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.4 Die Castings.-The die castings shall be of uniform quality free of cold shuts, blowholes and shrinkage defects. The castings must be clean and free from all chips. All unfilled castings shall be rejected.

3.4.1 Flash Removal. -All flash must be completely removed from parting lines, internal and external profiles, cored holes and ejector pin marks, except as noted on drawings.

3.4.2 Gates.-All waste material remaining from the gate must be removed except when so noted on the drawings.

3.4.3 Surface Porosity. -Surface porosity shall be limited to not more than one pore .020" x .050" x .020" deep max. on any area .250" square. Surfaces defined as any surface visible prior to sectioning.

3.4.4 Remelted Scrap. -No scrap shall be used other than sprews, gates, and rejected die castings of the particular alloy required by the item print and produced in the manufacturer's own plant, when that alloy is zinc. No aluminum scrap of any kind shall be used. The rejected die castings used shall be free from finishes, lubricants or any substance which may contaminate the alloy.

3.5 Hardness of Brass Ring. -The hardness of the brass ring shall be from Rockwell B55 to B70.

3.6 Workmanship.-All parts shall be fabricated and finished in a through, workmanlike manner, and all manufacturing, processing and assembly operations shall be correctly performed. The parts shall be clean and free of burrs, sharp edges, unblended radii, surface defects, cracks, chips, dirt, grease and oil (except where specifically required), rust and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat, legible and sharply defined.

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QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections.-The following types of inspection shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First Article Inspection

4.3.1 Submission.-The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities indicated.

a. Twenty-five (25) each of every component part

b. twenty-five (25) each of the subassemblies depicted on drawings 9207563 and 9211836

The first article sample shall consist of parts and assemblies which have been produced by the contractor or furnished by a supplier, and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract, and submit a record of this inspection with the sample, including statements of findings for materials, processes and tests. When multiple cavity mold is used in the fabrication of die cast components, a minimum of ten components form each mold cavity shall be included in the first article sample; 25 are required if only cavity is used. Each component shall bear a mark which identifies it with its particular cavity. The mark shall be located where subsequent machining shall not obliterate the mark. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer. A first article sample, or portion thereof as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or when a change occurs in manufacturing process, material used, drawing or specification such as to significantly effect product uniformity, as determined by the Government.

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4.3.2 Inspection to be Performed.-The first article subjected by the Government to any or all of the applicable examinations and tests specified in 4.4.2 and 4.4.3 of this specification and to any or all of the requirement of the applicable drawings and specifications.

4.3.3 Rejection.-See MIL-A-48078

4.4 Quality Conformance Inspections

4.4.1 Inspection Lot Formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination.-See MIL-A-48078

a. Sampling Plans.-Unless otherwise specified in the Classification Of Defects and Test Tables, sampling plans for Major and Minor Defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH 4.4.2.1	TITLE Ring, Time Train		SHEET 01 OF 01		DRAWING NUMBER 9207552
					NEXT HIGHER ASSEMBLY 9207554
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major A					
101	Depth of powder groove		0.40%	3.1	Gage
102	Width of powder groove		0.40%	3.1	Gage
103	Location of small hole from center		0.40%	3.1	Gage
104	Location of ends of groove from small hole		0.40%	3.1	Gage
105	Diameter of outside of powder groove		0.40%	3.1	Gage
106	Concentricity of outside of powder groove diameter with inside diameter		0.40%	3.1	Gage
107	Hardness	125		3.5	4.5.2
Major B					
131	Thickness		0.40%	3.1	Gage
132	Inside diameter		0.40%	3.1	Gage
133	Diameter of pin hole		0.40%	3.1	Gage
134	Foreign matter present in groove		0.40%	3.1	Visual
Minor					
201	Outside diameter		0.65%	3.1	Gage
202	Concentricity of inside diameter with outside diameter		0.65%	3.1	Gage
203	Burr present on any surface		0.65%	3.1	Visual
204	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.2	Ring Expelling Charge Retainer		01 01		9207551
					NEXT HIGHER ASSEMBLY 9207568
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major B					
131	Pitch diameter, min.		0.40%	3,1	Gage
132	Major diameter, max		0.40%	3.1	Gage
Minor					
201	Thickness		0.65%	3.1	Gage
202	Presence of two lugs		0.65%	3.1	Visual
203	Evidence of poor workmanship		0.65%	3.6	Visual

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	TITLE	SHEET OF	DRAWING NUMBER
4.4.2.3	Head	01 01	9207567
			NEXT HIGHER ASSEMBLY 9207568

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major A					
101	Location of shear wire hole		0.40%	3.1	Gage
Major B					
131	Diameter of safety wire hole, max.		0.40%	3.1	Gage
132	Location of safety wire hole		0.40%	3.1	Gage
133	Pitch diameter, min		0.40%	3.1	Gage
134	Major diameter, min.		0.40%	3.1	Gage
135	Distance from nose end to forward end of threaded Cavity		0.40%	3.1	Gage
136	Total length		0.40%	3.1	Gage
137	Surface porosity	80		3.4.3	4.5.3
Minor					
201	Distance from nose end to forward end of striker cavity, min.		0.65%	3.1	Gage
202	Diameter of shear wire hole		0.65%	3.1	Gage
203	Thickness through bottom or striker cavity		0.65%	3.1	Gage
204	Width of seating surface, max.		0.65%	3.1	Gage
205	Outside diameter of base, max.		0.65%	3.1	Gage
206	Distance between wrench slots, max.		0.65%	3.1	Gage
207	Evidence of poor workmanship		0.65%	3.6	Visual

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.4	Body		01 02		9207557
					NEXT HIGHER ASSEMBLY 9211836
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL 0R 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major A					
101	Location of pellet charge cavity		0.40%	3.1	Gage
102	Location of primer flash hole		0.40%	3.1	Gage
103	Location of pin hole		0.40%	3.1	Gage
104	Time train ring undercut diameter		0.40%	3.1	Gage
105	Depth of time train ring seat		0.40%	3.1	Gage
106	Pellet flash hole missing or obstructed		0.40%	3.1	Visual
107	Primer flash hole missing or obstructed		0.40%	3.1	Visual
108	Surface porosity	80		3.4.3	4.5.3
Major B					
131	Diameter primer cavity		0.40%	3.1	Gage
132	Pitch diameter of head threads, max.		0.40%	3.1	Gage
133	Major diameter of head threads, max.		0.40%	3.1	Gage
134	Pitch diameter, of fuze threads, max.		0.40%	3.1	Gage
135	Major diameter, of fuze threads, max.		0.40%	3.1	Gage
136	Concentricity of head threads with fuze threads		0.40%	3.1	Gage
137	Concentricity of outside diameter with fuze threads		0.40%	3.1	Gage
138	Depth of primer cavity		0.40%	3.1	Gage
139	Depth of pellet charge cavity		0.40%	3.1	Gage
140	Diameter of pellet charge cavity		0.40%	3.1	Gage
141	Pitch diameter of retaining ring thread, min.		0.40%	3.1	Gage

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH 4.4.2.4	TITLE		SHEET 02 OF 02		DRAWING NUMBER / NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARARGAPH REFERENCE /INSPECTION METHOD
142	Major diameter of retaining ring thread, min.		0.40%	3.1	Gage
143	Diameter of pin hole		0.40%	3.1	Gage
144	Depth of expelling charge cavity		0.40%	3.1	Gage
145	Diameter of expelling charge cavity, min.		0.40%	3.1	Gage
146	Diameter of shoulder for seating head		0.40%	3.1	Gage
Minor					
201	Outside diameter		0.65%	3.1	Gage
202	Distance across wrench slots		0.65%	3.1	Gage
203	Evidence of poor workmanship		0.65%	3.6	Visual

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.5	Pin		01 01		9207555
					NEXT HIGHER ASSEMBLY 9211836
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major A					
101	Diameter		0.40%	3,1	Gage
Minor					
201	Finish missing		0.65%	3.1	Visual
202	Evidence of poor workmanship		0.65%	3.6	Visual

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.6	Metal Parts Body Assembly		01 01		9211836
					NEXT HIGHER ASSEMBLY 9207556
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major A 101	Presence of pin		0.40%	3.1	Visual
Major B 131	Height of pin, maximum		0.40%	3.1	Gage
Minor 201	Evidence of poor workmanship		0.65%	3.6	Visual

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.7	Striker		01 01		9207564
					NEXT HIGHER ASSEMBLY 9207563
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major A					
101	Profile of point		0.40%	3.1	Gage
Major B					
131	Location of shear wire hole		0.40%	3.1	Gage
132	Location of safety wire hole		0.40%	3.1	Gage
133	Diameter of safety wire hole, max.		0.40%	3.1	Gage
Minor					
201	Total length		0.65%	3.1	Gage
202	Diameter of shear wire hole		0.65%	3.1	Gage
203	Protective finish missing		0.65%	3.1	Visual
204	Evidence of poor workmanship		0.65%	3.6	Visual

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	TITLE	SHEET OF	DRAWING NUMBER
4.4.2.8	Wire Shear	01 01	9207556
			NEXT HIGHER ASSEMBLY 9207563

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major A					
101	Diameter		0.40%	3,1	Gage
102	Shear wire load	125		3.2	4.5.1
Major					
201	Evidence of poor workmanship		0.65%	3.6	Gage

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.9	Wire Safety		01 01		9207565
					NEXT HIGHER ASSEMBLY 9207563
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Minor					
201	Protective finish missing or incomplete		0.65%	3.1	Visual
202	Evidence of poor workmanship		0.65%	3.6	Visual

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.10	Head Assembly	01	01	9207563
				NEXT HIGHER ASSEMBLY 9207568

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL 0R 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major A					
101	Shear wire not free		0.40%	3.1	Manual
102	Safety wire missing or broken		0.40%	3.1	Visual
103	Shear wire missing or broken		0.40%	3.1	Visual
104	Firing pin damaged		0.40%	3.1	Visual
105	Striker missing		0.40%	3.1	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.11	Disk Expelling Charge Retainer		01 01		9207560
					NEXT HIGHER ASSEMBLY 9207556
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL 0R 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major A					
101	Any hole missing		0.40%	3.1	Visual
Major B					
131	Outside diameter, min.		0.40%	3.1	Gage
Minor					
201	Thickness		0.65%	3.1	Gage
202	Evidence of poor workmanship		0.65%	3.6	Visual

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.12	Washer, Vent Crimping Support		01 01		9207550
					NEXT HIGHER ASSEMBLY 9207554
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL 0R 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major A 101	Inside diameter,		0.40%	3.1	Gage
Major B 131	Outside diameter		0.40%	3.1	Gage
Minor 201	Evidence of poor workmanship		0.65%	3.6	Visual

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4.4.3 Testing

4.4.3.1 Shear Wire Load Test

4.4.3.1.1 Beginning with the first lot produced and continuing until three(3) consecutive lots have been found acceptable, two hundred (200) shear wires shall be randomly selected from each lot for the test.

4.4.3.1.2 After three consecutive lots have been met the criteria specified in 4.4.3.1.1, the sample size shall be as specified in 4.4.2.8.

4.4.4 Inspection Equipment.-The inspection equipment required to perform the examinations and tests prescribed herein is described in 'Paragraph Reference/Investigation Method' column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Test Methods and Procedures

4.5.1 Shear Wire Load Test.-A hardened steel test head (dwg. 9207567) and hardened steel test striker (dwg. 9207564) shall be fabricated for this test, however the top of the nose of the head wire shall be left off to expose the striker cavity. The shear wire shall then be inserted through the head and striker shear wire openings with the ends of the wire turned two full turns so that the wire is loose and enables the striker to exhibit play in the head. Each wire in the sample shall be tested by applying a continuous and increasing load at a rate of not less than .10 nor more than .13 inch per minute to a punch. Note and record the load at shearing to determine compliance with the requirements of the test. If any shear wire fails to meet the requirements, the lot shall be rejected.

4.5.2 Hardness of Time Train.-For each time train ring tested two Rockwell B hardness readings shall be taken in the area enclosed by the projections of the ends of the black powder groove. A distance of at least 3/16 of an inch shall be maintained between readings. If two or more time train rings fail to meet the requirements of the test, the lot shall be rejected.

4.5.3 Surface Porosity.-The sample shall be inspected for surface porosity per the applicable requirements. If six or more samples fail to meet the requirements the lot shall be rejected.

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5. PREPARATION FOR DELIVERY

5.1 Packaging and Packing

5.1.1 Level C.-The parts shall be packaged and packed in accordance with Specification MIL-STD-1169.

5.2 Marking. The cartons shall be marked in accordance with Specification MIL-STD-129.

6. NOTES

6.1 Intended Use.-The parts and assemblies covered by this specification are intended for use on 60MM Mortar Ammunition.

6.2 Ordering Data.-See MIL-A-48078

6.3 Submission of Designs for Approval.-See MIL-A-48078. Submit equipment design, as required, to Commander, U.S. Army ARDEC, ATTN: SMCAR-QAT-B, Picatinny Arsenal, N. J. 07806-5000.

6.4 Submission of First Article Samples.-Instruction as to the location of the evaluation of the first article samples shall be obtained from the Contracting Officer. Upon receipt of such request, the Contracting Officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All Inquiries shall be forwarded to: Commander, U.S. Army ARDEC, ATTN: SMCAR-QAT-B, Picatinny Arsenal, N. J. 07806-5000.

Custodian:
ARMY-AR

Preparing Activity:
ARMY-AR

Project Number: 1390-A-071

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